

2.1 Introduction of Vector-Valued Functions

Definition: A vector-valued function or vector function, is simply a function whose domain is a subset of the real numbers and whose range is a set of vectors. If the vector function $\mathbf{r}(t)$ maps real numbers to n -dimensional vectors, we write: $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$.

Note: In this class we are most interested in vector functions of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$.

Note: Very strictly speaking, it is a slight abuse to write $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$ when the domain of $\mathbf{r}$ is not the full set of real numbers. In particular if D is the domain of $\mathbf{r}$ and it is a subset of the real numbers but not equal to the real numbers, we should write $\mathbf{r}: D \rightarrow \mathbb{R}^n$. We will ignore this subtlety in this class and be aware that $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$ only means that the domain of $\mathbf{r}$ is a subset of real numbers.

Definition: If a vector function is of the form $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then $f(t)$, $g(t)$, and $h(t)$ are called the component functions of $\mathbf{r}$.

Note: In this case we have $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$ because the real number t gets mapped to a vector $\langle f(t), g(t), h(t) \rangle$.

Example 1: Find the domain of the vector function: $\mathbf{r}(t) = \langle \sqrt{4-t^2}, \cos(t), \sqrt{t^2-1} \rangle$.

The domain of the first component function is all t in $[-2, 2]$.

The domain of the second component function is all real numbers.

The domain of the third component function is all t in $(-\infty, -1] \cup [1, \infty)$.

The domain of $\mathbf{r}$ will be all the values of t that are in the domain of all three component functions because each component function must spit something out if $\mathbf{r}$ is to spit out a vector. So, we take the intersection of all three domains above and we discover that the domain is all t in $[-2, -1] \cup [1, 2]$.

Note: It is important to see that as we graph a vector-valued function in this class we are simply graphing a parametric function in 3-space. Therefore, t will act as a parameter and each vector-valued function will have an associated orientation.

Example 2: We have seen that the parametric equations for the line that passes through the points $(2,-3,4)$ and $(4,2,1)$ are: $x = 2 + 2t$ $y = -3 + 5t$ $z = 4 - 3t$. How could one represent this line as a vector function?

A vector function for this line would be: $\mathbf{r}(t) = \langle 2 + 2t, -3 + 5t, 4 - 3t \rangle$.

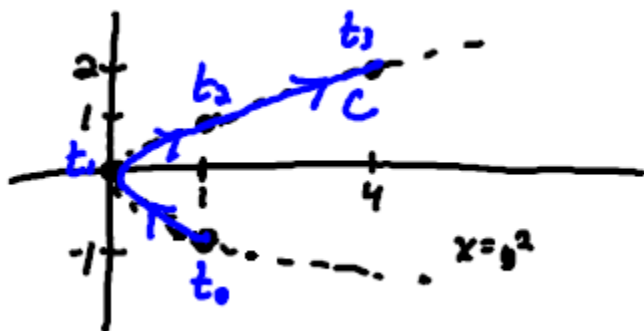
The instructor will show a computerized graph of this line.

Vector functions of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$ are very useful in physics for describing the motion of a particle that moves in a complicated fashion. In this context we usually think of t as representing time, and the position of the particle at time t is the terminal point of the position vector that the function spits out at t . Technology can help us to appreciate the power in this.

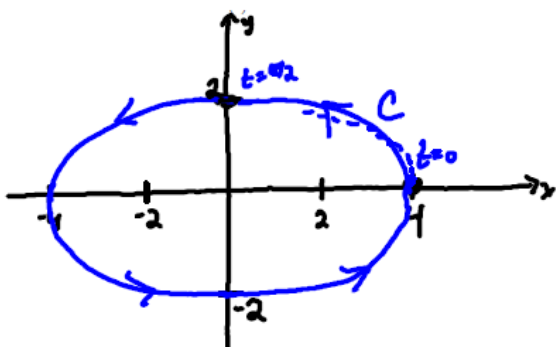
Example 3: Sketch a graph of each of the following functions.

a) $\mathbf{r}(t) = (t^2 - 2t + 1)\mathbf{i} + (t - 1)\mathbf{j}$ for $0 \leq t \leq 3$.

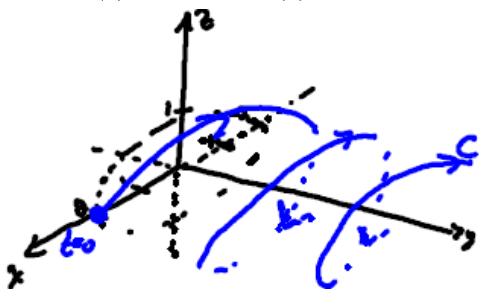
We may immediately note that $t^2 - 2t + 1 = (t - 1)^2$. Therefore we have the equation $x = y^2$.



b) $\mathbf{r}(t) = 4\cos(t)\mathbf{i} + 2\sin(t)\mathbf{j}$ for $0 \leq t \leq 2\pi$.



c) $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + 2t\mathbf{j} + \sin(t)\mathbf{k}$ for $t \geq 0$.



Example 4: Use a computer to graph the vector function:

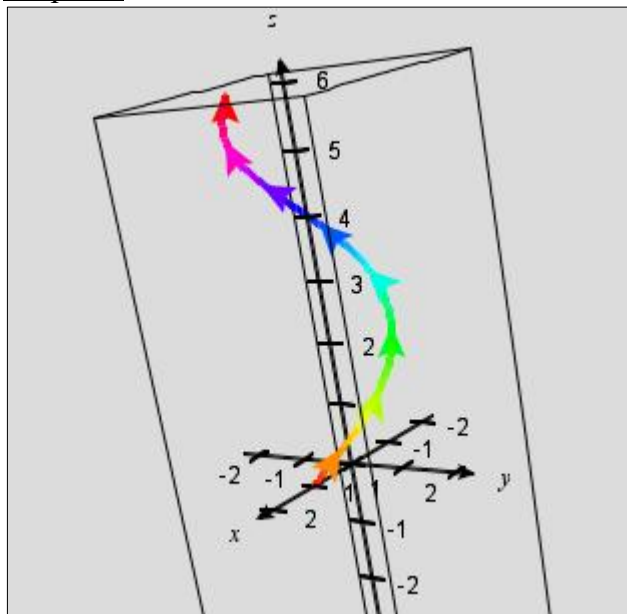
$$\mathbf{r}(t) = \langle (4 + \sin(20t))\cos(t), (4 + \sin(20t))\sin(t), \cos(20t) \rangle$$

One should use the free three dimensional plotter (the link is available in Blackboard) to do this. Bounds on t from -3.14 to 3.14 and 200 steps works well. Interestingly, this curve is called a toroidal spiral because it lies on a torus.

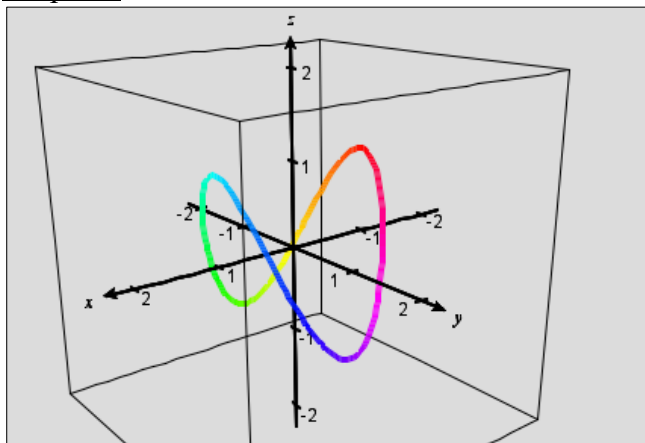
Note: One skill that a student should work to develop is the ability to match a curve with a vector function. The next example addresses this.

Example 5: Match each picture below with its appropriate vector function (you are not allowed to use a computer).

Graph A:



Graph B:



Possible functions:

1. $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$
2. $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t) \rangle$
3. $\mathbf{r}(t) = \langle \cos(t), t, \sin(t) \rangle$
4. $\mathbf{r}(t) = \langle \cos(t), \cos(t), \cos(t) \rangle$

Graph A should be matched with 1. Graph B should be matched with 2.